

Homework 7

Solution

1. Determine whether each of the following series converges or diverges:

(a)

$$\sum_{n=1}^{\infty} \frac{4^{3n} (n!)^2}{(2n+1)^{2n}}$$

Solution:

We use the ratio test

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{\left[4^{3n+3} \frac{((n+1)!)^2}{(2n+3)^{2n+2}} \right]}{4^{3n} \left[\frac{(n!)^2}{(2n+1)^{2n}} \right]} = \lim_{n \rightarrow \infty} \frac{64 \cdot (2n+1)^{2n} \cdot ((n+1)!)^2}{(2n+3)^{2n+2} \cdot (n!)^2} \\ &= \lim_{n \rightarrow \infty} \left[\frac{64 \cdot (n+1)^2}{(2n+3)^2} \cdot \left(\frac{2n+1}{2n+3} \right)^{2n} \right] = \lim_{n \rightarrow \infty} \left[\frac{64 \cdot \left(1 + \frac{1}{n}\right)^2}{\left(2 + \frac{3}{n}\right)^2} \cdot \frac{\left(1 + \frac{1}{2n}\right)^{2n}}{\left(1 + \frac{3}{2n}\right)^{2n}} \right] \stackrel{\text{AOL}}{=} \frac{64 \cdot e}{4 \cdot e^3} = \frac{16}{e^2} > 1 \end{aligned}$$

It follows that the series is divergent.

(b)

$$\sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdot 6 \cdots 2n}{n^n}$$

Solution:

We use the ratio test

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{\left[\frac{2 \cdot 4 \cdot 6 \cdots 2n \cdot (2n+2)}{(n+1)^{n+1}} \right]}{\left[\frac{2 \cdot 4 \cdot 6 \cdots 2n}{n^n} \right]} = \lim_{n \rightarrow \infty} \frac{(2n+2) \cdot n^n}{(n+1)^{n+1}} = \lim_{n \rightarrow \infty} 2 \cdot \left(\frac{n}{n+1} \right)^n \\ &= \lim_{n \rightarrow \infty} 2 \cdot \left(\frac{n+1}{n} \right)^{-n} = \lim_{n \rightarrow \infty} 2 \cdot \left(1 + \frac{1}{n} \right)^{-n} = \frac{2}{e} < 1 \end{aligned}$$

It follows that the series is convergent.

(c)

$$\sum_{n=1}^{\infty} \frac{(n!)^2}{2 \cdot 4 \cdot 6 \cdot \dots \cdot (4n)}$$

Solution:

We use the ratio test

$$L = \lim_{n \rightarrow \infty} \frac{a_{n+1}}{a_n} = \lim_{n \rightarrow \infty} \frac{[(n+1)!]^2 \cdot 2 \cdot 4 \cdot 6 \cdot \dots \cdot (4n)}{2 \cdot 4 \cdot 6 \cdot \dots \cdot (4n) \cdot (4n+2)(4n+4) \cdot (n!)^2} = \lim_{n \rightarrow \infty} \frac{(n+1)^2}{(4n+2)(4n+4)} = \lim_{n \rightarrow \infty} \frac{(1+\frac{1}{n})^2}{(4+\frac{2}{n})(4+\frac{4}{n})} = \frac{1}{16}$$

It follows that $L < 1$ and consequently, the series is convergent.

(d)

$$\sum_{n=1}^{\infty} \left[n^{2n} \cdot \left(1 - \cos\left(\frac{1}{n}\right) \right)^n \right]$$

Solution:

We use the root test

$$\begin{aligned} L &= \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \left[n^2 \cdot \left(1 - \cos\left(\frac{1}{n}\right) \right) \right] = \lim_{n \rightarrow \infty} \frac{1 - \cos\left(\frac{1}{n}\right)}{\left(\frac{1}{n^2}\right)} \stackrel{\text{Heine}}{=} \lim_{x \rightarrow 0} \frac{1 - \cos(x)}{x^2} \\ &= \lim_{x \rightarrow 0} \left[-\frac{\cos(x) - 1}{x^2} \right]^{-\frac{1}{2}} \stackrel{\text{famous}}{=} \frac{1}{2} < 1 \end{aligned}$$

Thus, the series is convergent.

(e)

$$\sum_{n=2}^{\infty} \frac{\sqrt[n]{n^2} \cdot a^n}{n^{1.5}}, \quad \text{where } a \geq 0$$

Solution:

We use the root test

$$L = \lim_{n \rightarrow \infty} \sqrt[n]{a_n} = \lim_{n \rightarrow \infty} \sqrt[n]{\frac{\sqrt[n]{n^2} \cdot a^n}{n^{1.5}}} = \lim_{n \rightarrow \infty} \frac{\sqrt[n]{n^2} \cdot a}{(\sqrt[n]{n})^{1.5}} = a$$

It follows that the series converges for $0 \leq a < 1$ and diverges for $a > 1$.For $a = 1$ the series is $\sum_{n=2}^{\infty} \frac{\sqrt[n]{n^2}}{n^{1.5}}$, and it converges with the 1.5-Harmonic series as

$$L = \lim_{n \rightarrow \infty} \frac{a_n}{b_n} = \lim_{n \rightarrow \infty} \frac{\left[\frac{\sqrt[n]{n^2}}{n^{1.5}} \right]}{\left[\frac{1}{n^{1.5}} \right]} = \lim_{n \rightarrow \infty} (\sqrt[n]{n})^2 = 1$$

Thus, the series is convergent if and only if $0 \leq a \leq 1$.

(f) Student A solved item (b) in the following way:

We compute

$$\lim_{n \rightarrow \infty} \frac{\sqrt[3]{2 \cdot 4 \cdot 6 \cdots 2n}}{\sqrt[n]{n^n}} = \lim_{n \rightarrow \infty} \frac{2^{\frac{1}{n}} \cdot 4^{\frac{1}{n}} \cdots (2n)^{\frac{1}{n}}}{n} \stackrel{\text{AOL}}{=} \frac{1 \cdot 1 \cdot 1 \cdots 1}{\infty} = 0$$

Thus, by the root test, the series $\sum_{n=1}^{\infty} \frac{2 \cdot 4 \cdot 6 \cdots 2n}{n^n}$ is convergent.

Explain student A's mistake.

Solution:

Student A abused the algebra of limits. He used the algebra of limits on a number of functions that is not fixed, but depends on n . However, when using AOL, one needs to determine the number of functions and only then use the theorem, independently of n .

2. Find all values of $p \geq 0$ for which the following series is convergent

$$\sum_{n=3}^{\infty} \frac{1}{n \cdot \ln(n) \cdot \ln(\ln(n)) \cdot [\ln(\ln(\ln(n)))]^p}$$

Solution:

Let $p \geq 0$ and consider the function defined by

$$f(x) = \frac{1}{x \cdot \ln(x) \cdot \ln(\ln(x)) \cdot [\ln(\ln(\ln(x)))]^p} \quad \forall x \geq 27$$

Note that

$$x \geq 27 = 3^3 \Rightarrow x > e^e \Rightarrow \ln(x) > e \Rightarrow \ln(\ln(x)) > 1 \Rightarrow \ln(\ln(\ln(x))) > 0$$

Thus, f is well defined.

We observe that since $p \geq 0$, we have $\lim_{x \rightarrow \infty} f(x) = \frac{1}{\infty} = 0$. Also, note that f is decreasing, since $x \cdot \ln(x) \cdot \ln(\ln(x)) \cdot [\ln(\ln(\ln(x)))]^p$ is increasing and positive for every $x \geq 27$. Thus, by the integral test, the series is convergent if and only if $\int_{27}^{\infty} f(x) dx$ is convergent. We compute

$$\begin{aligned} \int_{27}^{\infty} f(x) dx &= \int_{27}^{\infty} \frac{dx}{x \cdot \ln(x) \cdot \ln(\ln(x)) \cdot [\ln(\ln(\ln(x)))]^p} \stackrel{t=\ln(x) \Rightarrow dx=x dt}{=} \int_{\ln(27)}^{\infty} \frac{dt}{t \cdot \ln(t) \cdot [\ln(\ln(t))]^p} \\ &\stackrel{u=\ln(t) \Rightarrow dt=t du}{=} \int_{\ln(\ln(27))}^{\infty} \frac{du}{u \cdot [\ln(u)]^p} \stackrel{s=\ln(u) \Rightarrow du=sd s}{=} \int_{\ln(\ln(\ln(27)))}^{\infty} \frac{ds}{s^p} \end{aligned}$$

Thus, the series is convergent if and only if $p > 1$.

3. In this problem we will prove the existence of the Euler constant, that will be re-visited in the course of Probability.

Let $(T_N)_{N=1}^\infty$ be the sequence of real numbers defined by $T_N = \sum_{n=1}^N \frac{1}{n} - \int_1^N \frac{dx}{x}$ for every $N \in \mathbb{N}$.

(a) Prove that $(T_N)_{N=1}^\infty$ is decreasing.

Hint. recall the inequality $f(n+1) \leq \int_n^{n+1} f(x) dx \leq f(n)$ that we have seen in lecture 6.

Solution:

We note that for every $n \in \mathbb{N}$ we have

$$\frac{1}{n+1} \underset{(\star\star)}{\leq} \int_n^{n+1} \frac{dx}{x} \underset{(\star)}{\leq} \frac{1}{n}$$

Let $N \in \mathbb{N}$. Then

$$T_{N+1} - T_N = \sum_{n=1}^{N+1} \frac{1}{n} - \int_1^{N+1} \frac{dx}{x} - \left(\sum_{n=1}^N \frac{1}{n} - \int_1^N \frac{dx}{x} \right) =$$

$$\frac{1}{N+1} - \int_1^{N+1} \frac{dx}{x} + \int_1^N \frac{dx}{x} = \frac{1}{N+1} - \int_N^{N+1} \frac{dx}{x} \underset{(\star\star)}{\leq} \frac{1}{N+1} - \frac{1}{N+1} = 0$$

Thus $(T_N)_{N=1}^\infty$ is decreasing.

(b) Conclude that there exists a constant $\gamma \geq 0$ for which $\gamma = \lim_{N \rightarrow \infty} \left[\sum_{n=1}^N \frac{1}{n} - \ln N \right]$.

Solution:

Let $N \in \mathbb{N}$. Then

$$T_N = \sum_{n=1}^N \frac{1}{n} - \int_1^N \frac{dx}{x} = \sum_{n=1}^N \left(\frac{1}{n} - \int_n^{n+1} \frac{dx}{x} \right) + \int_N^{N+1} \frac{dx}{x} \underset{(\star)}{\geq} 0 + \int_N^{N+1} \frac{dx}{x} \geq 0$$

thus $(T_N)_{N=1}^\infty$ is bounded from below by 0 and therefore $\lim_{N \rightarrow \infty} T_N$ exists, and by the monotonicity of limits we get that

$$\lim_{N \rightarrow \infty} T_N \geq \lim_{N \rightarrow \infty} 0 = 0$$

We note that for every $N \in \mathbb{N}$ we have

$$\int_1^N \frac{dx}{x} = [\ln |x|]_1^N = \ln N - \ln 1 = \ln N$$

thus there exists an $\gamma \geq 0$ for which

$$\gamma = \lim_{N \rightarrow \infty} \left[\sum_{n=1}^N \frac{1}{n} - \ln(N) \right]$$

4. Prove or disprove each of the following statements:

(a) Let $(a_n)_{n=1}^{\infty}$ be a sequence of non-negative numbers.

If $\lim_{n \rightarrow \infty} (n \cdot a_n) = 0$, then $\sum_{n=1}^{\infty} a_n < \infty$.

Solution:

False.

We choose $a_1 = 1$ and $a_n = \frac{1}{n \cdot \ln n}$ for every $2 \leq n \in \mathbb{N}$.

We note that

$$\lim_{n \rightarrow \infty} (n \cdot a_n) = \lim_{n \rightarrow \infty} \left[n \cdot \frac{1}{n \cdot \ln n} \right] = \lim_{n \rightarrow \infty} \frac{1}{\ln n} = 0$$

but we proved in class that

$$\sum_{n=2}^{\infty} \frac{1}{n \cdot \ln n} = \infty$$

Thus $\sum_{n=1}^{\infty} a_n = \infty$, since it has a divergent tail.

(b) Let $(a_n)_{n=1}^{\infty}$ be a sequence of positive numbers. Suppose that

$$\frac{a_{n+1}}{a_n} \leq \frac{n^2}{(n+1)^2} \quad \forall n \in \mathbb{N}$$

then the series $\sum_{n=1}^{\infty} a_n$ is convergent.

Solution:

True.

Consider the sequence $(b_n)_{n=1}^{\infty}$ defined by $b_n = \frac{1}{n^2}$ for every $n \in \mathbb{N}$.

Note that for every $n \in \mathbb{N}$ we have

$$\frac{a_{n+1}}{a_n} \leq \frac{n^2}{(n+1)^2} = \frac{\left\lfloor \frac{1}{(n+1)^2} \right\rfloor}{\left\lfloor \frac{1}{n^2} \right\rfloor} = \frac{b_{n+1}}{b_n}$$

Since the 2-Harmonic series is convergent, the series $\sum_{n=1}^{\infty} a_n$ is convergent.

(c) We have

$$\frac{\pi}{2} \leq \sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}} \leq \frac{2\pi}{3}$$

You can assume that $\pi > 3$.

Solution:

True.

Consider the function defined by

$$f(x) = \frac{1}{(x+1)\sqrt{x}}, \quad \forall x \geq 1$$

We note that $f(x) > 0$ for every $x \geq 1$.

We note that f is decreasing (as the functions $x+1, \sqrt{x}$ are increasing and positive for every $x \geq 1$)

and that $\lim_{x \rightarrow \infty} f(x) \stackrel{\text{AOL}}{=} 0$.

We now compute the improper integral

$$\int_1^{\infty} \frac{dx}{(x+1)\sqrt{x}} \stackrel{t=\sqrt{x}}{=} \int_1^{\infty} \frac{2t \, dt}{(t^2+1)t} = 2 \int_1^{\infty} \frac{dt}{1+t^2} = 2 [\arctan(t)]_1^{\infty} = 2 \left[\frac{\pi}{2} - \frac{\pi}{4} \right] = \frac{\pi}{2}$$

Thus, by the integral test, we have

$$\int_1^{\infty} \frac{dx}{(x+1)\sqrt{x}} \leq \sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}} \leq f(1) + \int_1^{\infty} \frac{dx}{(x+1)\sqrt{x}}$$

Therefore

$$\frac{\pi}{2} \leq \sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}} \leq \frac{1}{2} + \frac{\pi}{2}$$

As $\pi > 3$, then

$$\frac{1}{2} \leq \frac{\pi}{6}$$

and therefore

$$\frac{1}{2} + \frac{\pi}{2} \leq \frac{\pi}{6} + \frac{\pi}{2} = \frac{2\pi}{3}$$

It follows that

$$\frac{\pi}{2} \leq \sum_{n=1}^{\infty} \frac{1}{(n+1)\sqrt{n}} \leq \frac{2\pi}{3}$$